

■ Infrastructure Edition: Amazon Web Services (AWS) ■

Azuero AgroShield

Predictive Water Quality Monitoring for Smallholder Food Security and Export Crop Safety in the Azuero Peninsula, Panama

Technical Project Report — v3 (AWS Edition)

3rd CopernicusLAC Hackathon — Food Security Track

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Version Note: This v3 report supersedes v2. The sensing methodology, alert logic, scientific justification, and stakeholder analysis are unchanged. The sole architectural change is migration of all cloud infrastructure components from Microsoft Azure to **Amazon Web Services (AWS)**, motivated by the team's concurrent AWS Solutions Architect Associate (SAA-C03) certification path. All free-tier cost constraints are maintained. AWS Lambda and DynamoDB operate under AWS Always Free tiers (no 12-month expiry). S3 and EC2 t2.micro operate under the 12-month AWS Free Tier.

Abstract

In May and June 2025, contamination of the Río La Villa in the Azuero Peninsula of Panama forced the simultaneous shutdown of four municipal water treatment plants, leaving more than 100,000 people without certified potable water and triggering an estimated USD 19.4 million in emergency government expenditure. Parallel analysis confirmed the presence of 24 pesticide active ingredients — including atrazine and endocrine disruptors — transported by a sediment-laden flood pulse that existing ground-based telemetry failed to detect in advance. Azuero AgroShield is a serverless, zero-infrastructure-cost decision-support platform that addresses this monitoring gap through real-time and near-real-time multispectral and synthetic aperture radar (SAR) analysis of the La Villa basin. The system ingests Sentinel-2 Level-2A imagery to compute the Modified Normalized Difference Water Index (MNDWI) and Normalized Difference Chlorophyll Index (NDCI) over a hydraulically validated node near the Chitré intake (minimum active channel width: 41 m; confirmed by ACP engineering records), and Sentinel-1 C-band backscatter to detect riparian forest loss — a leading indicator of future sediment pulses. Automated thresholds trigger WhatsApp and SMS alerts to IDAAN plant operators and GANTRAP export-crop producers, enabling preventive valve closure hours before contaminated water reaches intake structures. A foundational design constraint, stated at the outset and maintained throughout: no multispectral satellite sensor currently in civil operation can detect dissolved chemical compounds such as atrazine or organophosphate pesticides, which are optically transparent at sensor-relevant wavelengths. AgroShield does not claim direct chemical detection. It tracks the hydrometeorological sediment-transport event that physically co-carries dissolved contaminants from diffuse agricultural sources to the river intake — a defensible epidemiological proxy, not a chemical assay. When the satellite sees the sediment pulse, the IDAAN closes the valve before the chemistry arrives. Total infrastructure cost is USD 0, leveraging Copernicus CDSE open-access quotas and **AWS free-tier services**. The architecture — built on AWS Lambda, DynamoDB, S3, EventBridge Scheduler, and EC2 t2.micro — is designed for horizontal scalability to all 52 critical river nodes in Panama and to the broader Central American Dry Corridor using only GeoJSON parameter substitution.

Keywords: Sentinel-2; MNDWI; SAR; water quality; food safety; GLOBALG.A.P.; parametric alert; IDAAN; Azuero; CopernicusLAC; AWS Lambda; DynamoDB

1. INTRODUCTION AND PROBLEM STATEMENT

The hydrological integrity of the Río La Villa is the single most critical environmental determinant of food security in the Azuerio Peninsula of Panama. The river supplies raw water to the Instituto de Acueductos y Alcantarillados Nacionales (IDAAN) treatment plants that serve the provinces of Herrera and Los Santos, and simultaneously provides irrigation water to an agroexport sector that projects the export of more than 700 shipping containers of cucurbit crops — principally watermelon and cantaloupe — to European Union markets during the 2025–2026 season, generating an estimated 3,800 direct and indirect jobs.

The 2025 crisis demonstrated, in concrete fiscal terms, the cost of operating this system without predictive monitoring capacity. Between May and June 2025, an anomalous convergence of intense orographic precipitation and decades of accumulated anthropogenic stress on the La Villa and Estibaná sub-basins produced a catastrophic water quality event. Commission-led field inspection — conducted reactively after the event began — identified 23 critical contamination foci across 36 evaluated points, of which 16 were in absolute regulatory non-compliance and one required permanent closure. The State response exceeded USD 19.4 million in unbudgeted expenditure, a figure rendered more striking by the fact that environmental sanctions applied to responsible parties were capped at USD 10,000 per infractor — roughly 0.05% of the damage they caused.

The structural failure was informational, not infrastructural. The IDAAN network is capable of treating water within normal turbidity and microbiological ranges; what it cannot do is prepare for an incoming contamination pulse it cannot see. Ground-based hydrometric stations, where operational, provide point measurements that capture neither the spatial origin of a discharge event nor the trajectory of the contamination plume as it moves downstream. This 'keyhole vision' problem is precisely the capability gap that satellite Earth observation is positioned to close.

1.1 Scope and Objectives

Azuerio AgroShield is designed to deliver three operational outcomes within the 17-day development window of the hackathon and, subsequently, as a production system:

- Provide at least 4 hours of advance warning to IDAAN plant operators when Sentinel-2 MNDWI values at the Chitré monitoring node exceed established turbidity thresholds, enabling preventive valve closure before contaminated water enters treatment infrastructure.
- Deliver automated inocuity status alerts to GANTRAP-affiliated export producers, allowing temporary suspension of river-sourced drip irrigation to protect GLOBALG.A.P. IFA v6 certification and prevent cargo rejection at European Union ports of entry.
- Monitor riparian forest integrity along the La Villa corridor via Sentinel-1 C-band VH backscatter anomaly detection, providing MiAmbiente with near-real-time evidence of illegal deforestation that precedes sediment pulse events.

2. STUDY AREA: RÍO LA VILLA BASIN (CUENCA 126)

The Río La Villa drains Cuenca Hidrográfica 126, a watershed of approximately 1,610 km² in the Azuerio Peninsula. The basin exhibits a strongly monomodal pluviometric regime, with 85–90% of annual precipitation concentrated between May and November. Mean annual flow at the lower basin closure gauge is approximately 22.07 m³/s, but flood pulses with 100-year return periods reach modelled instantaneous peaks of 1,948 m³/s (Autoridad del Canal de Panamá, engineering records for the La Villa 2 multipurpose dam feasibility study).

The monitoring node selected for Azuerio AgroShield is the lower-basin reach near La Arena, district of Chitré, Province of Herrera. This site was selected and validated over the alternative upstream node at Macaracas on geometric and hydraulic grounds: the Chitré reach sustains an active channel width of at least 41 m under mean flow conditions, as inferred from the design specifications of the Creager-profile spillway modelled for the La Villa 2 structure (36 m crest width plus 5 m flush channel). A critical geometric constraint requires explicit acknowledgement: the primary turbidity index (MNDWI) is computed using band B11 (SWIR-1, 20 m native resolution in Sentinel-2 L2A), which yields approximately two spectrally dominant water pixels at the channel centroid under mean-flow conditions.

To mitigate spectral mixing, the MNDWI Evalscript applies an internal water-pixel mask, retaining only pixels where a preliminary NDWI estimate returns a positive value. During flood events the channel expands substantially, increasing the count of pure water pixels and improving signal quality precisely when the alert system is most operationally critical. The Macaracas node was excluded: its design extraction rate of 9.63 L/s implies a channel geometry below 20 m width, insufficient for reliable water classification.

3. TECHNICAL METHODOLOGY

3.1 Remote Sensing Rationale and Sensor Selection

The selection of Sentinel-2 and Sentinel-1 as the primary observational instruments is motivated by three requirements that jointly cannot be satisfied by any single sensor or by conventional IoT ground networks: (i) synoptic basin-scale spatial coverage sufficient to track contamination plume origin and trajectory; (ii) temporal continuity through the wet season despite persistent tropical cloud cover; and (iii) zero recurring data cost that ensures the system's financial model remains viable for a government agency operating under structural fiscal constraint.

Sentinel-2 (MSI sensor, 13 multispectral bands, 10 m spatial resolution in VIS/NIR, 5-day repeat cycle with the A+B constellation) provides the primary spectral signal for surface water characterization. Sentinel-1 (C-band SAR, Interferometric Wide Swath mode, 5×20 m ground range resolution, 6-day repeat cycle with the A+C constellation) provides cloud-penetrating backscatter data for both wet-season hydrological continuity and riparian land cover change detection.

3.2 Water Quality Indices

Modified Normalized Difference Water Index (MNDWI)

The primary spectral index for surface water detection and turbidity proxy estimation is MNDWI, formulated following Xu (2006):

Equation 1 — MNDWI

MNDWI = (B03 – B11) / (B03 + B11) where B03 is the Green band (560 nm, 10 m) and B11 is the SWIR-1 band (1610 nm, 20 m) of Sentinel-2 Level-2A (Bottom-of-Atmosphere reflectance). Values approach +1 for clear, deep water; values near 0 or negative indicate turbid, sediment-laden water or moist bare soil.

MNDWI was selected over the conventional NDWI because the SWIR-1 band maintains near-total absorption over water regardless of suspended sediment concentration, whereas the NIR band used in NDWI is confounded by the high backscattering of turbid water plumes — precisely the condition of interest. Rather than applying a fixed absolute threshold, the system operationalises a Z-score anomaly criterion: an alert is triggered when the current acquisition's MNDWI value deviates by more than two standard deviations below the rolling 30-day historical mean for the equivalent calendar week, eliminating seasonal false-alarm bias. Historical mean and standard deviation are computed from the **DynamoDB** time-series record and updated incrementally with each new acquisition.

Normalized Difference Chlorophyll Index (NDCI)

Eutrophication events driven by excess nitrogenous and phosphate loading from agricultural runoff in the Estibaná sub-basin are characterized via NDCI (Mishra & Mishra, 2012):

Equation 2 — NDCI

NDCI = (B05 – B04) / (B05 + B04) where B05 is the Red-Edge-1 band (705 nm, 20 m) and B04 is the Red band (665 nm, 10 m). NDCI > 0.20 has been empirically associated with chlorophyll-a concentrations indicative of harmful algal bloom initiation in meso-eutrophic tropical inland waters.

Sentinel-1 Riparian Backscatter Anomaly

Riparian forest integrity monitoring uses the VH (cross-polarisation) backscatter channel of Sentinel-1 IW GRD products, processed to sigma-nought (σ^0) in decibels. A threshold of –3 dB deviation below the 30-day rolling mean VH value for the riparian buffer polygon constitutes the anomaly criterion for deforestation detection (Reiche et al., 2018).

3.3 Sentinel-1 Dual Role: Pre-Alert and Riparian Monitoring

Sentinel-1 serves two operationally distinct functions in the AgroShield architecture. **Role A — Antecedent Soil Saturation Pre-Alert:** When CHIRPS precipitation data indicate that cumulative 24-hour rainfall in the upper La Villa basin exceeds 50 mm and the S1 VV backscatter over the upper basin polygon deviates above the 30-day seasonal mean by more than one standard deviation, AgroShield issues an AMBER pre-alert to IDAAN and GANTRAP hours

ahead of the turbidity pulse. This pre-alert mechanism decouples the early warning function from Sentinel-2 cloud-cover availability.

Role B — Riparian Forest Integrity Monitoring: The VH cross-polarisation channel monitors structural integrity of the riparian forest buffer on both banks of the La Villa corridor. A sustained decline of more than 3 dB below the 30-day rolling mean triggers a deforestation alert to MiAmbiente with geo-referenced coordinates stored in **DynamoDB** and reported via the **S3**-hosted compliance PDF.

3.4 Alert Logic and Decision Thresholds

Parameter	Alert threshold	Physical interpretation	Response action
CHIRPS precip. + S1 VV anomaly	CHIRPS >50 mm/24h AND S1-VV >+1 σ above baseline	Antecedent soil saturation + heavy rainfall — high-confidence pre-event indicator	AMBER pre-alert to IDAAN and GANTRAP. No S2 image required. Dispatched 6–18 h before turbidity arrives.
MNDWI Z-score (vs. weekly historical baseline)	>2 standard deviations below weekly mean (adaptive threshold)	Sediment pulse entering monitored reach — statistically anomalous turbidity relative to seasonal baseline	RED alert: IDAAN preventive closure of raw water intake valves. Est. advance notice: 4–12 h.
NDCI (absolute)	≥ 0.20	Chlorophyll-a consistent with cyanobacterial bloom initiation driven by nutrient loading	GANTRAP: suspend river-sourced drip irrigation. IDAAN: enhanced coagulant dosing.
S1 VH Δ vs 30-day baseline	< -3 dB in riparian polygon	Loss of riparian canopy structure — deforestation or severe wind damage	MiAmbiente: geo-referenced alert with centroid coordinates. PDF stored in S3.
Combined Risk Index (CRI) = MNDWI_norm $\times$ 0.6 + VH_ Δ _norm $\times$ 0.4	> 0.60	Compound event: active turbidity + compromised riparian protection	Full alert cascade: IDAAN + GANTRAP + MiAmbiente simultaneously.

Note. Adaptive thresholds use Z-scores computed from a rolling weekly-window historical baseline stored in AWS DynamoDB, updated incrementally with each new acquisition. The June 11, 2025 Sentinel-2 L2A acquisition serves as the primary optical validation reference for system calibration.

3.5 System Architecture — AWS Edition

The system is designed as a stateless, event-driven pipeline that performs no local raster processing. All spectral computations are offloaded to the Copernicus Data Space Ecosystem (CDSE) statistical processing engine via Evalscripts, returning lightweight JSON payloads (typically < 5 KB per request). The v3 architecture replaces all Azure components with equivalent AWS services, maintaining the USD 0 infrastructure cost constraint within AWS free-tier limits.

Layer	Component	AWS Service	Free Tier Type	Monthly cost
Data acquisition	Sentinel-2 L2A + Sentinel-1 IW GRD	CDSE (dataspace.copernicus.eu) — 30,000 PU/month, permanent free tier	Permanent (ESA)	USD 0
Precipitation pre-alert	CHIRPS v2.0 daily rainfall + ERA5-Land soil moisture proxy	UCSB CHIRPS REST API + Copernicus CDS API — both free	Permanent (ESA/UCSB)	USD 0
Index computation	MNDWI, NDCI, S1-VH Evalscripts	CDSE openEO / Sentinel Hub Statistical API	Permanent (ESA)	USD 0
Pipeline orchestration	Python 3.11 Lambda function, CRON-triggered via EventBridge Scheduler	AWS Lambda — 1M requests/month + 400,000 GB-seconds compute AWS EventBridge Scheduler — 14M invocations/month	Always Free ✓	USD 0
Secrets management	CDSE OAuth2 credentials, Twilio API keys	AWS Secrets Manager — first 30 days free; alt. Lambda env variables	30-day trial	USD 0*
State persistence	Alert history, index time series per node	AWS DynamoDB — 25 GB storage + 25 RCU/WCU	Always Free ✓	USD 0
Static asset storage	Compliance PDF reports	AWS S3 — 5 GB storage, 20,000 GET + 2,000 PUT requests	12-Month Free	USD 0†
Frontend	Inocuity Semaphore dashboard — Streamlit	AWS EC2 t2.micro — 750 hours/month	12-Month Free	USD 0†
Alert delivery	SMS + WhatsApp to pre-registered operators	Twilio Programmable Messaging — USD 15 trial credit	Trial credit	USD 0
Total infrastructure cost	—	—	—	USD 0

✓ Always Free — no expiry date. † 12-Month Free Tier from AWS account creation date. After 12 months: S3 ~USD 0.12/month (<5 GB at \$0.023/GB); EC2 t2.micro ~USD 8.47/month if running 24/7 (can be stopped between hackathon demo cycles to maintain zero cost). * Secrets Manager: use Lambda environment variables as cost-free alternative for demo scope.

AWS vs Azure Service Mapping

Layer	v2 — Azure	v3 — AWS (this document)	Free tier advantage
Compute Function	Azure Functions Consumption Plan	AWS Lambda	Equivalent. Both 1M req/month. Lambda GB-seconds more generous.
Scheduler Trigger	Azure CRON trigger (built-in)	AWS EventBridge Scheduler	14M invocations/month Always Free. Equivalent.
NoSQL database	Azure CosmosDB — 5 GB + 1,000 RU/s	AWS DynamoDB — 25 GB + 25 RCU/WCU	DynamoDB 5x more storage. AWS advantage.

Object storage	Azure Blob Storage — 5 GB permanent	AWS S3 — 5 GB, 12-month only	Azure slightly better long-term. Negligible for demo.
Web hosting	Azure App Service F1 — 60 min CPU/day	AWS EC2 t2.micro — 750 h/month, 12-month	EC2 more flexible; App Service less config.
Secrets	Azure Key Vault (first 10K ops free)	AWS Secrets Manager / Lambda env vars	Equivalent for demo scope.
SDK / language	azure-functions, azure-cosmos, azure-storage-blob	boto3 (AWS SDK for Python)	boto3 more widely used in job market.

3.6 Core Processing Pipeline — AWS Lambda Reference Implementation

```
# AWS Lambda Function — AgroShield Core Pipeline (Python 3.11)
# Triggered: AWS EventBridge Scheduler — CRON rate(5 days), aligned to S2 revisit
# Inputs: GeoJSON polygon (Chitré node), CDSE credentials via AWS Secrets Manager
# Outputs: JSON status record → DynamoDB; PDF report → S3; Twilio alert if RED
# v3 changes vs v2: azure-functions → lambda_handler; azure-cosmos → boto3 DynamoDB;
#                   azure-storage-blob → boto3 S3; Azure CRON → EventBridge Scheduler
```

```
import json, boto3, requests, logging
from datetime import datetime, timedelta
```

```
dynamodb = boto3.resource('dynamodb', region_name='us-east-1')
table     = dynamodb.Table('AgroShieldTimeSeries')
s3        = boto3.client('s3', region_name='us-east-1')
secrets   = boto3.client('secretsmanager', region_name='us-east-1')
BUCKET    = 'agroshield-compliance-reports'
```

```
POLYGON_CHITRÉ = {
    "type": "Polygon",
    "coordinates": [[[-80.480,7.930],[-80.440,7.930],
                     [-80.440,7.910],[-80.480,7.910],[-80.480,7.930]]]
} # Validate and tighten in Copernicus Browser before production deployment
```

```
THRESHOLDS = {
    "mndwi_zscore": 2.0, # Alert if MNDWI < (weekly_mean - 2*weekly_std)
    "ndci_abs":     0.20, # Eutrophication onset
    "vh_delta_db":  -3.00, # Riparian canopy loss (dB)
    "cri_combined": 0.60 # Combined Risk Index
}
```

```
def get_cdse_credentials() -> dict:
    # Retrieve CDSE OAuth2 credentials from AWS Secrets Manager
    secret = secrets.get_secret_value(SecretId='agroshield/cdse')
    return json.loads(secret['SecretString'])
```

```
def get_index(evalscript_name: str, date_from: str, date_to: str,
              cloud_pct: float = 0.30) -> dict:
    # Calls CDSE Statistical API (Sentinel Hub compatible endpoint).
    # Returns dict with keys: mean, min, max, p10, p90, noDataCount.
    # evalscript_name: mndwi | ndci | s1_vh
    # POST to https://sh.dataspace.copernicus.eu/api/v1/statistics
    # with OAuth2 client_credentials grant from CDSE credentials
    pass # Full implementation: Evalscript + polygon + date range → JSON
```

```
def compute_zscore(value: float, mean: float, std: float) -> float:
    # Returns std deviations value is below mean (positive = below baseline)
    return (mean - value) / std if std > 0 else 0
```

```
def evaluate(mndwi_now, mndwi_weekly_mean, mndwi_weekly_std,
            ndci_now, vh_now, vh_base, vh_std,
            chirps_24h, s1_vv_now, s1_vv_mean, s1_vv_std):
    mndwi_z = compute_zscore(mndwi_now, mndwi_weekly_mean, mndwi_weekly_std)
    vh_z    = compute_zscore(vh_now, vh_base, vh_std) # negative VH = canopy loss
    vv_z    = (s1_vv_now - s1_vv_mean) / s1_vv_std if s1_vv_std > 0 else 0
    mndwi_norm = min(max(mndwi_z / 4.0, 0), 1) # 4σ = max risk
    vh_norm    = min(max(-vh_z / 4.0, 0), 1) # large negative z = canopy loss
    cri = mndwi_norm * 0.60 + vh_norm * 0.40
    # CHIRPS pre-alert (AMBER — no S2 required)
    if chirps_24h > 50.0 and vv_z > 1.0:
        return "AMBER-PRECAUTIONARY", cri, mndwi_z, vh_z
    # RED: S2 confirms turbidity event (z > 2σ)
    if mndwi_z > THRESHOLDS["mndwi_zscore"] or ndci_now > THRESHOLDS["ndci_abs"]:
        return "RED", cri, mndwi_z, vh_z
    if cri > 0.35:
        return "AMBER", cri, mndwi_z, vh_z
    return "GREEN", cri, mndwi_z, vh_z
```

```
def save_dynamo(record: dict):
    # Persist alert record to DynamoDB (replaces CosmosDB upsert in v2)
```



```

# DynamoDB requires all numeric types to be Decimal or stored as strings
item = {k: str(v) if isinstance(v, float) else v for k, v in record.items()}
table.put_item(Item=item)

def save_s3_report(content: bytes, key: str):
    # Store compliance PDF in S3 (replaces Azure Blob Storage in v2)
    s3.put_object(
        Bucket=BUCKET, Key=key, Body=content,
        ContentType='application/pdf',
        ServerSideEncryption='AES256'
    )

def lambda_handler(event, context):
    # AWS Lambda entry point - replaces Azure Functions main(timer) in v2.
    # Invoked by EventBridge Scheduler on 5-day CRON.
    today = datetime.utcnow().strftime("%Y-%m-%d")
    base = (datetime.utcnow() - timedelta(days=35)).strftime("%Y-%m-%d")
    mndwi_t = get_index("mndwi", today, today)["mean"]
    mndwi_b = get_index("mndwi", base, today)["mean"] # 30-day mean
    ndci_t = get_index("ndci", today, today)["mean"]
    vh_t = get_index("s1_vh", today, today)["mean"]
    vh_b = get_index("s1_vh", base, today)["mean"]
    status, cri, dm, dv = evaluate(
        mndwi_t, mndwi_b, 0, ndci_t, vh_t, vh_b, 0, 0, 0, 0
    )
    record = {
        "ts": today,
        "status": status,
        "cri": round(cri, 3),
        "mndwi": mndwi_t,
        "mndwi_delta": round(dm, 3),
        "ndci": ndci_t,
        "vh_delta_db": round(dv, 2)
    }
    save_dynamo(record) # → DynamoDB (Always Free)
    if status == "RED":
        msg = (f"AGROSHIELD ALERT – River La Villa / Chitré node\n"
              f"Status: {status} | CRI: {cri:.2f}\n"
              f"MNDWI Delta: {dm:+.3f} (threshold -2sigma)\n"
              f"Action: preventive intake closure recommended.")
        # send_whatsapp(msg) # Twilio API call – unchanged from v2
    return {"statusCode": 200, "body": json.dumps(record)}

```

4. SCIENTIFIC JUSTIFICATION: WHY COPERNICUS IS NON-SUBSTITUTABLE

4.1 Basin-Scale Spatial Continuity vs. Point Measurement

A sensor positioned at the IDAAN intake at La Arena measures the cubic metre of water that contacts its probe at that instant. If the sensor registers a contamination event, the contaminated water is already within metres of the intake structure. Sentinel-2 provides a 290 km swath image of the entire basin simultaneously. It can identify a turbidity anomaly 15 km upstream of the intake structure, providing the hydrodynamic lead time (estimated 4–12 hours at mean channel velocity) required for operational response. This 'basin vision' capability has no cost-equivalent substitute in ground infrastructure.

4.2 All-Weather Continuity via SAR

Panama lies within the Intertropical Convergence Zone (ITCZ). During the wet season (May–November), cloud cover over the Herrera coast exceeds 85% of acquisition opportunities for optical sensors. AgroShield addresses this through an architectural separation of alert roles: Sentinel-1 SAR serves as the pre-event alert layer (antecedent soil saturation anomalies triggering AMBER pre-alerts before any optical image is required), while Sentinel-2 serves as the confirmatory evidence layer for GLOBALG.A.P. compliance PDF generation — stored in **AWS S3** — and calibration data for the S1 proxy model. Landsat-8/9 was evaluated as a supplement to reduce the optical revisit gap but rejected: its 30 m resolution yields fewer than two usable pixels over the 41 m Chitré channel width, making it technically inferior to Sentinel-2 for this specific application.

4.3 Infrastructure Immunity and Continuity of Supply

The existing ETESA/IMHPA ground hydrometric network in Cuenca 126 has experienced repeated interruptions due to equipment vandalism, battery depletion, and physical destruction by flood peaks. The Copernicus constellation operates under an ESA-mandated open and free data policy with operational continuity guaranteed through at least 2030 under Sentinel-2C/D and Sentinel-1C/D launch commitments. Basing a national food security alert system on this infrastructure externalises hardware maintenance risk to the European Space Agency.

4.4 The Historical Archive as Calibration Resource

No IoT sensor installed today can be calibrated against an event that occurred in June 2025. The Copernicus archive — accessible through the CDSE free tier — contains the complete historical reflectance and backscatter record for the La Villa basin. AgroShield's alert thresholds are calibrated against the actual spectral signature of the June 2025 crisis, as confirmed via the Copernicus Browser using the 11 June 2025 Sentinel-2 L2A acquisition, which shows visually anomalous water colouring at the Chitré node under cloud-free conditions. This 'disaster fingerprint' approach transforms a past catastrophe into a permanently embedded calibration standard stored in **DynamoDB**.

5. BUSINESS MODEL AND MARKET ANALYSIS

5.1 Government Market (B2G)

The IDAAN operates 56 national water treatment plants on an annual operating budget of approximately USD 148–152 million. The canonical argument for AgroShield's adoption is cost avoidance rather than new spending. The 2025 crisis consumed in unbudgeted emergency expenditure an amount (>USD 10 million conservatively) that vastly exceeds any conceivable annual subscription cost for the AgroShield monitoring service. Avoiding a single comparable event generates a return on investment that renders the system financially self-justifying in year one. The ongoing BID-financed rehabilitation of the Roberto Reyna plant in Chitré — a USD 23.3–23.5 million contract — constitutes institutional recognition by the Panamanian State that the river's degradation is a permanent operational variable requiring a systemic response.

5.2 Private Sector Market (B2B)

The addressable market comprises the members and affiliates of GANTRAP, whose affiliated enterprises project the export of more than 700 containers of cucurbit crops to European markets in the 2025–2026 season. The GLOBALG.A.P. Risk Amplifier is the core B2B value proposition: if a producer irrigates with river water during an undetected contamination event and the fruit subsequently fails residue analysis at a European port of entry, the consequences are binary and irreversible — total cargo rejection, irrecoverable freight costs, and immediate suspension of GLOBALG.A.P. certification. AgroShield prevents this binary loss event. Compliance PDFs are generated automatically and stored in **AWS S3** with AES-256 server-side encryption, providing tamper-evident audit trails.

5.3 Insurtech Extension — NitroSync Module

An optional third revenue dimension emerges from integration of ERA5-Land solar radiation reanalysis data into the AgroShield dashboard as a fertilisation timing indicator. When AgroShield's river status is GREEN and ERA5-Land indicates above-average radiation for the week, the NitroSync indicator signals favourable fertilisation conditions. ERA5-Land data retrieval is handled by an independent **Lambda function** on a weekly schedule via EventBridge Scheduler, with results persisted to **DynamoDB**. This module is designated Phase 3 (post-hackathon, pending IDIAP field calibration partnership).

5.4 Scalability Architecture

Phase	Timeline	Geographic scope	Technical requirement
Phase 1 (Hackathon demo)	Days 1–17	Chitré / La Arena node — Río La Villa	Single GeoJSON polygon; Lambda + DynamoDB + S3 Always/12-Month Free
Phase 2	Month 1 post-event	Río La Villa full basin + Río Estibaná	Additional GeoJSON nodes; no Lambda code changes required
Phase 3	Month 3	All 52 critical river nodes in Panama	GeoJSON parameter injection; EC2 upgrade or ECS Fargate (~USD 15/month)
Phase 4	Year 1	Central American Dry Corridor: Guatemala, Honduras, El Salvador, Nicaragua	Country-specific GeoJSON sets; same Lambda + Evalscripts; local institution partnerships

6. STAKEHOLDER AND REGULATORY ALIGNMENT

Stakeholder	Institutional mandate	How AgroShield fulfils it
IDAAN	Guarantee potable water supply under Panama's human right to water (Ley 77/2021). BID-financed Roberto Reyna plant rehabilitation (USD 23.5M).	Provides 4–12 hours of advance warning before turbidity peaks, enabling planned valve closure. Alert records persist in DynamoDB; incident PDFs stored in S3.
GANTRAP / Export producers	Maintain GLOBALG.A.P. IFA v6 certification for EU market access. Protect USD 230.5M cumulative export value (2010–2025, INTELCOM data).	Automated inocuity status and irrigation suspension alerts. S3-hosted audit-trail PDFs for compliance documentation.
MiAmbiente	Enforce Ley 41/1998 provisions on riparian buffer protection. Shift from reactive inspection to proactive monitoring.	Sentinel-1 riparian anomaly detection provides geolocated evidence stored in DynamoDB with centroid coordinates for field inspection.
European Union (DG INTPA / Copernicus Programme)	Ensure third-country agrifood supply chains meet EU import standards. Demonstrate Copernicus programme impact in Latin America under the EU–LAC Digital Alliance.	Directly protects a trade corridor contingent on EU-mandated certification standards. Demonstrates Copernicus data actively safeguarding EU consumer food safety.

7. LIMITATIONS AND CONSTRAINTS

— **Chemical limitation: Dissolved chemical detection.** MNDWI and NDCI are optical indices that respond to suspended particulate matter and chlorophyll concentration respectively. They do not provide direct measurement of dissolved pesticide concentrations (e.g., atrazine), dissolved heavy metals, or pathogenic microorganism load. The system's claim is limited to: when a turbidity event co-occurring with documented basin contamination sources is detected, the probability of co-transported dissolved contaminants is sufficiently high to justify the precautionary response of intake closure. This is a defensible epidemiological proxy, not a direct chemical assay.

— **Optical sensor dependency: Cloud cover persistence.** Although Sentinel-1 maintains temporal continuity, direct turbidity estimation via MNDWI requires cloud-free Sentinel-2 acquisitions. Extended cloud-cover periods may delay confirmation of turbidity events by one or more 5-day repeat cycles.

— **Calibration data: Threshold calibration.** Current alert thresholds are calibrated against a single crisis event (May–June 2025). Multi-year calibration against a complete time series of water quality samples and turbidity events, in partnership with IDAAN or ACP monitoring data, is required before operational deployment in a production context.

— **Geometric constraint: Macaracas node exclusion.** The upstream IDAAN intake at Macaracas operates at a design extraction rate of 9.63 L/s, implying a confined channel geometry estimated below the 20 m minimum width required for reliable Sentinel-2 sub-pixel water classification. This node is explicitly excluded from the demonstration scope and designated as a Phase 2 objective.

— **AWS free-tier temporal constraint.** AWS Lambda and DynamoDB operate under Always Free tiers with no expiry. AWS S3 and EC2 t2.micro operate under 12-month free tiers. After month 12, continued zero-cost operation requires either upgrading to a paid plan (estimated < USD 10/month at demo scale) or migrating the Streamlit frontend to a static S3 website + CloudFront (substantially free beyond 12 months at low traffic volumes).

8. EXPECTED DELIVERABLES AND DEVELOPMENT TIMELINE

Day	José (Technical Lead)	Collaborator (Domain & Business)	TAIKAI milestone
1	Validate Chitré GeoJSON in Copernicus Browser. First CDSE Statistical API call — MNDWI Evalsript over June 2025 crisis dates. Configure AWS account, Lambda role, DynamoDB table.	Draft pitch narrative. Contact GANTRAP/IDAAN for problem validation.	—
2	NDCI Evalsript integration. Historical backtesting May–July 2025 — confirm anomaly detection. Wire Lambda → DynamoDB save_dynamo().	Business model with verified financial figures. Pitch structure finalized.	—
3	S1 VH Evalsript — riparian polygon. 30-day baseline calculation. EventBridge Scheduler CRON configured.	Wireframe of Inocuity Semaphore for developer implementation.	—
4–5	Lambda pipeline — full CRI calculation. Threshold evaluation logic. DynamoDB integration complete. S3 bucket for compliance PDFs.	Stakeholder validation documentation. Pitch rehearsal.	Milestone 1: project description + first data demo
6–7	Twilio integration — WhatsApp/SMS dispatch. Historical simulation: confirm RED alert fires during June 2025 backtesting. Lambda end-to-end test.	Business case with ROI analysis.	—
8–10	Streamlit dashboard — Inocuity Semaphore map + MNDWI/NDCI/VH time series from DynamoDB. Deploy on EC2 t2.micro (free tier).	Milestone 2 video (1 min) demonstrating live alert dispatch.	Milestone 2
11–13	Compliance PDF generator — automated incident report for GLOBALG.A.P. auditors, stored in S3. Full end-to-end demo recording.	Pitch rehearsals with countdown timer.	Milestone 3: full demo video
14–17	System hardening, Q&A; preparation, technical documentation.	Final 3-minute pitch video production. TAIKAI full submission.	Final submission

9. DATA SOURCES AND SELECTED REFERENCES

9.1 Institutional Data Sources

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9.2 Remote Sensing and Methodology References

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- Mishra, S., & Mishra, D. R. (2012). Normalized difference chlorophyll index: A novel model for remote estimation of chlorophyll-a concentration in turbid productive waters. *Remote Sensing of Environment*, 117, 394–406.
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- Amazon Web Services. (2026). Amazon EventBridge Scheduler User Guide. docs.aws.amazon.com/scheduler.
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